

Instructions

What is structured expert elicitation?

Structured expert elicitation is a method for obtaining judgements from experts that minimises bias and may reflect uncertainty. It is used to obtain beliefs from experts in a quantitative or statistical form, for example, as probability distributions for uncertain quantities.

What types of questions will we ask you?

We will ask you to estimate some of our model inputs, such as average appointment length or the average proportion of USC (urgent suspected cancer) referrals that result in a cancer diagnosis. We are interested in your uncertainty around these values.

What do we mean by uncertainty?

Uncertainty refers to how (un)sure you are about an answer to a question. For example, suppose we ask for the proportion of patients that get diagnosed with cancer following a USC referral. You might judge your best guess to be 9%. You might also judge that it's very unlikely to be below 6% or above 12%. This is uncertainty! It describes a range of plausible answers to a question so that you can provide your degree of belief about a single quantity.

This is different to thinking about variation between individuals. For example, for a referral clinic in location A, only 4% of referrals may result in a cancer diagnosis, while for a referral clinic in location B, 14% of referrals may results in a diagnosis. This isn't uncertainty because it refers to variation between different clinics, instead of a range of plausible values for the same type of or average referral clinic.

How will you express your uncertainty?

In order to express your uncertainty, first you will be asked to give a range of possible values. For example, if we are asking you for the average proportion of USC referrals that result in a cancer diagnosis, the plausible range represents

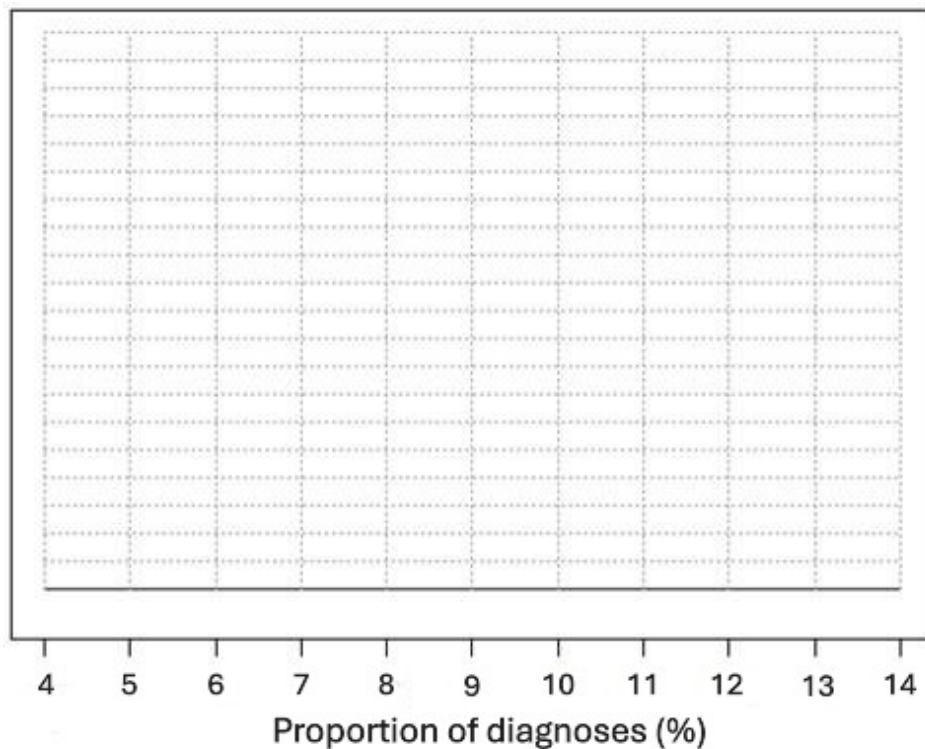
- Your lowest plausible proportion of diagnoses (minimum) - a value such that you believe that there is a 1% probability that the proportion of diagnoses is less than that value.
- Your highest plausible proportion of diagnoses (maximum) - a value, such that you believe that there is a 1% probability that the proportion of diagnoses is more than that value.
- So you believe that there is 98% probability between the lowest and the highest values.

Test your range by imagining that somebody gives a value that is outside your plausible range. Your reaction should be that the person has misunderstood or misremembered, i.e. you are very confident that you have chosen the right range!

Let's say that the average proportion of diagnoses is unlikely to be less than 5, or more than 13 percent.

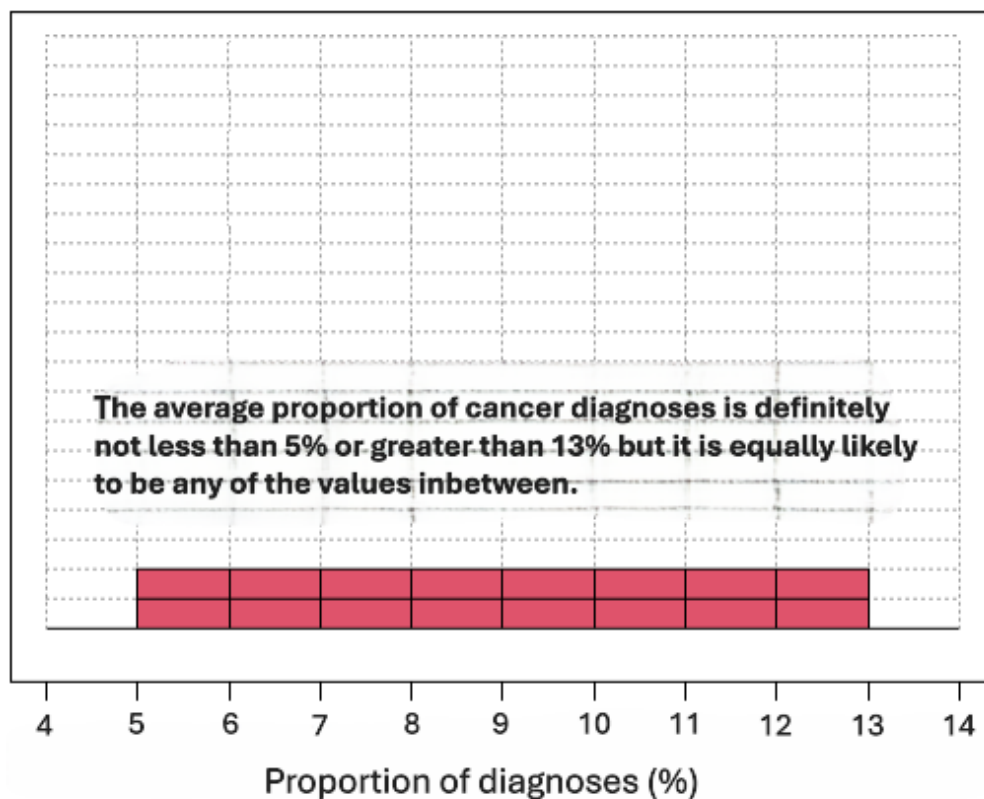
After giving your plausible range, you will then be required to fill in a histogram, like the one below. The range of possible values that appear along the bottom are determined by the range you specify - in this case 4% to 14% (slightly wider than the specified range of 5% to 13%). Note that the grid is split into columns (we call these bins) - the example below has ten bins (4-5%, 5-6%, 6-7%, and so on).

To tell us how plausible you think each of these values in this range are, you will be given a number of chips to place in the bins. There are a different number of *chips* depending on the range you give.

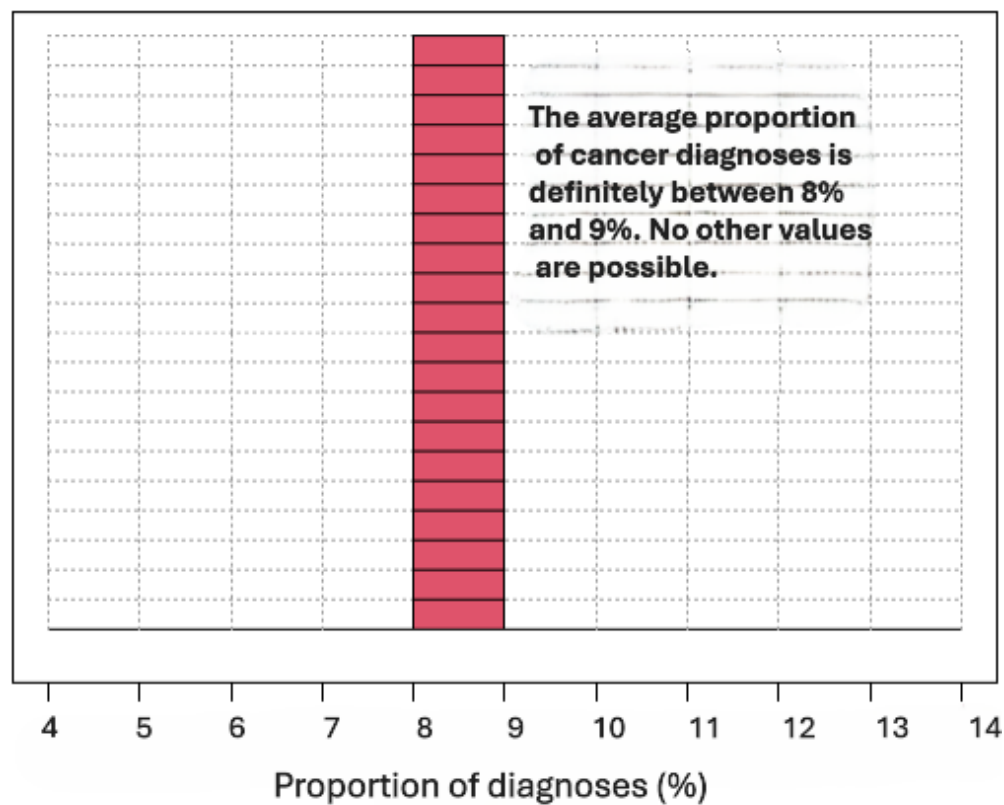


Use this to indicate how confident you feel that each of the values on the horizontal axis is the correct average value. The higher the red bar in the grid, the more confident you are.

For example, if you are completely uncertain, you place the same number of chips in each of the bins, so the height of red bars is the same for all possible values, as shown below.

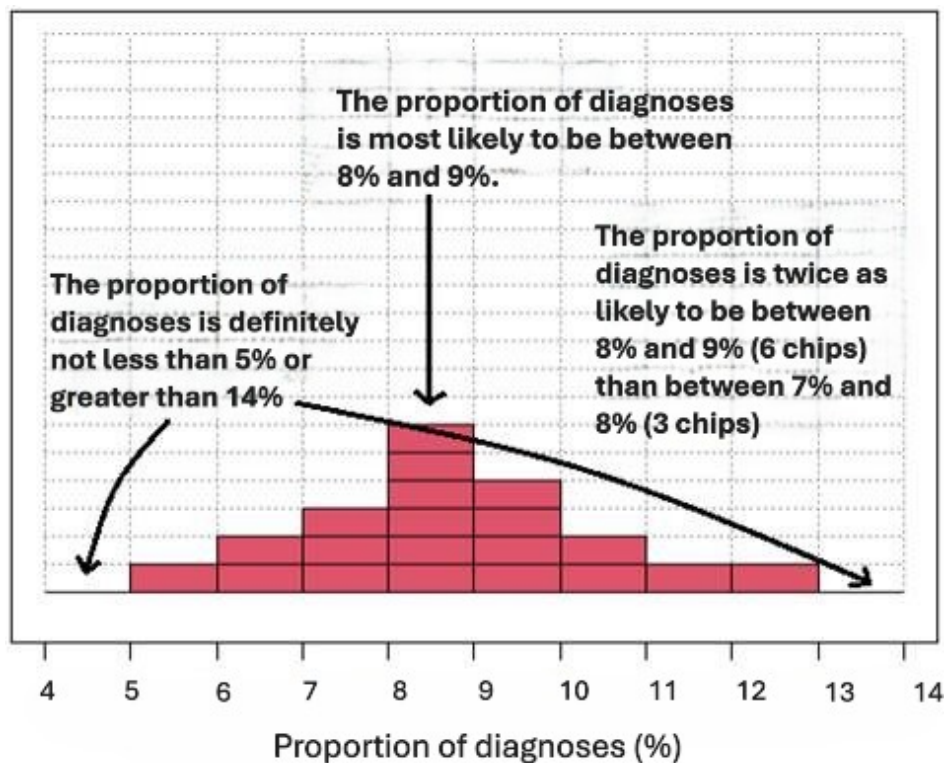


Similarly, if you are absolutely certain of the value, you place all chips in one bin, as shown below.

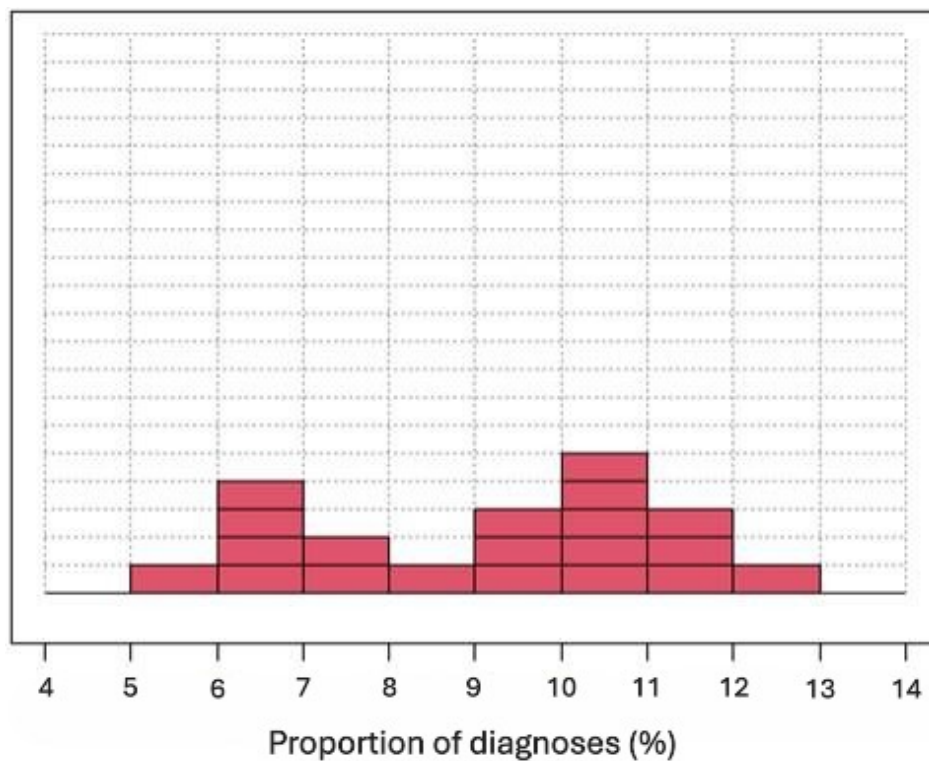


You will need to be mindful about what your response says about the degree of certainty in your belief. For example, twice as many coloured boxes on one value means you believe that the value is twice as likely to be the answer.

Another example is shown below.



It's important to realise that, because we are asking you about the average and the uncertainty around this, there is no rationale for breaks in the bins, like in the grid below.

**Be aware of bias!**

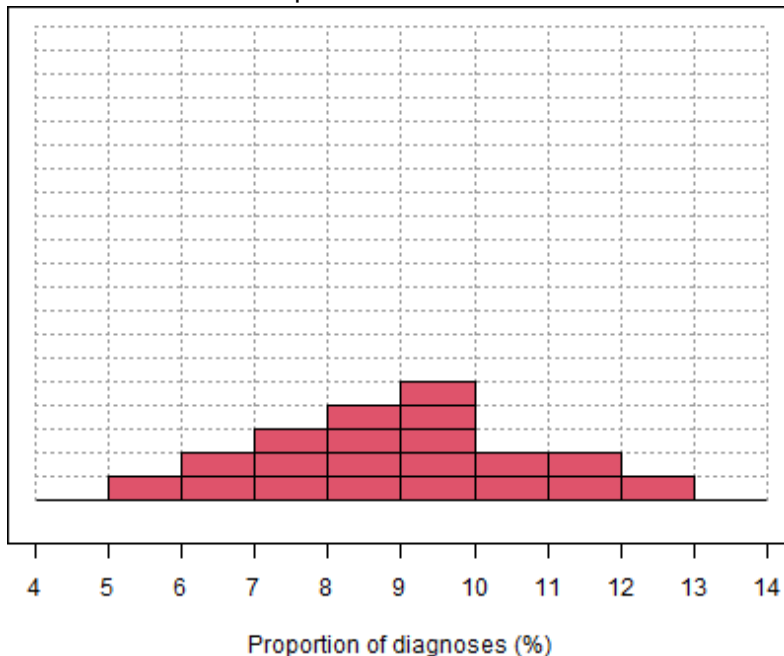
Cognitive biases such as fixating on an initial value or relying on easily recalled examples can distort estimates, while motivational biases like favouring information that confirms existing beliefs, wishful thinking or precautionary thinking may skew judgments towards preferred outcomes. These often result in overconfidence, producing distributions that are too narrow. Being aware of these biases during elicitation helps reduce their impact.

You can practice with the grid below:

What proportion of USC (urgent suspected cancer) referrals for patients with pre-existing anxiety and/or depression result in a cancer diagnosis?

Please add 20 chips to the grid below to express your uncertainty. The more chips you place in a particular bin the more certain you are that the proportion lies in that bin.

You can use **0** more chips.



Summary

Your answers imply that

- There is a 50% probability that the proportion of cancer diagnoses from USC referrals, on average, is between 5 and 9 %
- There is a 25% probability that the proportion of cancer diagnoses from USC referrals, on average, is between 9 and 10 %
- There is a 25% probability that the proportion of cancer diagnoses from USC referrals, on average, is between 10 and 13 %

If these summary statements do not represent your beliefs you can modify the grid.

Once you are satisfied that you can express your beliefs in this way, click on 'Next' to continue.

Next

Note you will be able to refer to these instructions at any point in the exercise by clicking on the 'Instructions' tab.